WEEK 12 OVERVIEW

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CEVE 543

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What Should We Trust in a Climate Model?

Weather Types

WHAT CAN A CLIMATE MODEL TELL YOU ABOUT DAILY RAINFALL IN 2070?

- Its rainfall at the nearest grid cell?
- Its winds and pressure patterns over South America?
- Its global mean temperature?

OHIO EXTREME RAIN DAYS

Note

Figure shown in class: Spring extreme-rain days per year (left) and their persistence (right), GFDL CM3 (black) and observed (red). Farnham et al. (2018), Fig. 3.

ON THOSE DAYS, THE MODEL'S CIRCULATION LOOKS LIKE REANALYSIS

Note

Figure shown in class: 700 hPa height anomalies on extreme-rain days: two GFDL CM3 runs and reanalysis. Farnham et al. (2018), Fig. 5.

TWO WAYS TO USE A CLIMATE MODEL

Trust its local rainfall

- Correct the distribution at each grid cell.
- The historical distribution matches observations by construction.
- Wet and dry spells, and year-to-year variance, come from the model.

Trust its large-scale circulation

- Take the *sequence of weather patterns* from the model.
- Model rainfall from large-scale weather patterns

OHIO EXTREME RAINFALL DAYS

Logistic regression for the chance of a regional extreme-rain day, given five daily atmospheric indices (pressure patterns, humidity, vertical motion); fit to observations, then driven by the model's indices.

i Note

Figure shown in class: Spring extreme-rain days from model rainfall (solid black), model circulation (dashed), and observations (red). Farnham et al. (2018), Fig. 8.

What Should We Trust in a Climate Model?

Weather Types

CAN WE CLUSTER CIRCULATIONS?

Note

Figure shown in class: Circulation (top) and rainfall (bottom) anomalies for six weather types over subtropical South America. Doss-Gollin et al. (2018), Fig. 6.

THE 2015/16 PARAGUAY RIVER FLOODS

Note

Figure shown in class: Daily rainfall over the lower Paraguay basin, NDJF 2015/16, labeled by weather type. Doss-Gollin et al. (2018), Fig. 7.

DO MODELS REPRODUCE WEATHER TYPE PATTERNS?

Note

Figure shown in class: Spring 500 hPa height anomalies by type (red high, blue low): reanalyses (top two rows), GFDL models. Muñoz et al. (2017), Fig. 3.

THIS WEEK

- Wednesday: discussion of Moron et al. (2008). Read the paper, bring 2 questions and paper only.
- Friday: Lab 8